

1) Encrypt "CHASE YOUR DREAMS"

a) Caesar Cipher ($n=3$)

$$C = (P+3) \bmod 26$$

Plaintext	$P+3$	$(P+3) \bmod 26$	Ciphertext
C	5	5	F
H	10	10	K
A	3	3	D
S	21	21	V
E	7	7	H
Y	27	1	B
O	17	17	R
U	23	23	X
R	20	20	U
D	6	6	G
R	20	20	U
E	7	7	H
A	3	3	D
M	15	15	P
S	21	21	V

b) Shift Cipher with Key 7

$$C = (P + 7) \bmod 26$$

Plain text $P+7$ $(P+7) \bmod 26$ Ciphertext

C	9	9	J
H	14	14	O
A	7	7	H
S	25	25	Z
E	11	11	L
Y	31	5	F
O	21	21	V
U	27	1	B
R	24	24	Y
D	10	10	K
R	24	24	Y
E	11	11	L
A	7	7	H
M	19	19	T
S	25	25	Z

Ciphertext: JOH ZL FVBY KYLHTZ

c) ROT 13

Plaintext $P+13$ $(P+13) \bmod 26$ Ciphertext

C	18	18	P
H	20	20	U
A	13	13	N
S	31	5	F
E	17	17	R
Y	37	11	L
O	27	1	B
U	33	7	H
R	30	4	E
D	16	16	Q
R	30	4	E
E	17	17	R
A	13	13	N
M	25	25	Z
S	31	5	F

Cipher Text = PUNFR LBHE QERNZF

2) Decrypt

a) "WRWDOOB VHFU PHVVDJH" using Caesar cipher

Ciphertext	$P(C-3)$	$(C-3) \bmod 26$	Plaintext
W = 22	19	19	T
R = 17	14	14	O
W = 22	19	19	T
D = 3	0	0	A
O = 14	11	11	L
O = 14	11	11	L
B = 1	-2	24	Y
V = 21	18	18	S
H = 7	4	4	E
F = 5	2	2	C
U = 20	17	17	R
H = 7	4	4	E
W = 22	19	19	T
P = 15	12	12	M
H = 7	4	4	E
V = 21	18	18	S
V = 21	18	18	S
D = 3	0	0	A
J = 9	6	6	G
H = 7	4	4	E

Plaintext = TOTALLY SECRET MESSAGE

b) "STB DTWP YNRJ" using Shift Cipher, 5

Ciphertext	C-5	$(C-5) \bmod 26$	Plaintext
S = 18	13	13	N = 14
T = 19	14	14	O = 15
B = 1	-4	22	W = 23
D = 3	-2	24	Y = 25
T = 19	14	14	O = 15
W = 22	17	17	R = 18
P = 15	10	10	K = 11
Y = 24	19	19	T = 20
N = 13	8	8	I = 9
R = 17	12	12	M = 13
I = 9	4	4	E = 5

Plaintext = NEW YORK TIME

"MJQQT BTWQI" using shift cipher, 5

Ciphertext	C-5	$(C-5) \bmod 26$	Plaintext
M = 12	7	7	H = 8
J = 9	4	4	E = 5
Q = 16	11	11	L = 12
Q = 16	11	11	L = 12
T = 19	14	14	O = 15
B = 1	-4	22	W = 23
T = 19	14	14	O = 15
W = 22	17	17	R = 18
Q = 16	11	11	L = 12
I = 8	3	3	D = 4

Plain text
HELLO
WORLD

c) "FRAQ ZBER PBSSRR" using ROT13

Ciphertext C- $(C-) \bmod 26$ Plaintext

F = 5 -8 18 S

R = 17 4 4 E

A = 0 -13 13 N

Q = 16 3 3 D

Z = 25 12 12 M

B = 1 -12 14 O

E = 4 -9 17 R

R = 17 4 4 E

P = 15 2 2 C

B = 1 -12 14 O

S = 18 5 5 F

S = 18 5 5 F

R = 17 4 4 E

R = 17 4 4 F

Plaintext = SEND MORE COFFEE